

Brian Xia

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EDUCATION

Cornell University Ithaca, New York
Bachelor of Arts, Computer Science (GPA: 3.70) Aug 2024 - May 2028

WORK EXPERIENCE

Prudential Financial Newark, NJ
Global Technology & Operations Intern Jun 2026 - Present

- Proposed and built a full-stack tool (React, Express) using mutation testing across 2 internal repos to surface unit tests with inflated coverage but weak fault detection, integrating Claude (Bedrock) to generate 30 targeted test cases that lifted test effectiveness by 25.6% per file
- Migrated microfrontend and BFF API calls from Apigee to Kong, updating auth headers and adapting to downstream OpenAPI spec changes as part of a company-wide gateway migration
- Pioneered and documented Playwright adoption for automated QA testing across microfrontend apps

Cornell University Ithaca, NY
Teaching Assistant, Object-Oriented Programming and Data Structures Jan 2026 - Present

- Led weekly discussions on data structures and algorithms (recursion, sorting, trees, graphs) for 21 students
- Hosted office hours to assist students with debugging Java programs and algorithmic reasoning
- Evaluated 60+ Java assignments and provided feedback on code quality, algorithm correctness, and OOP design

CYNC Cambridge, MA (Remote)
Software Development Intern August 2025 - Dec 2025

- Built AWS Lambda microservices and GraphQL resolvers for CYNC's wellness platform, processing wearable health data to generate personalized health recommendations
- Implemented authentication flows with AWS Cognito, supporting standard email/password login and OAuth 2.0
- Developed React Native mobile app features to enable real-time syncing of wearable data

J2 Lab - Cornell University Ithaca, NY
Undergraduate Researcher Feb 2025 - Dec 2025

- Developed neural network models achieving over 97% accuracy using meta-learning (TensorFlow, BoTorch) to optimize superconductor film production and reduce experimental time and cost
- Optimized model training with tensor computations, achieving a 36x reduction in training time for faster experiments

PROJECTS

WellScout (Nonprofit Project) Jun 2025 - Aug 2025

- Shipped a full-stack platform for water well site planning across regions, incorporating environmental risk and accessibility analysis in consultation with nonprofit field experts
- Migrated client-side fetches to Next.js API routes, reducing query timeouts for large-dataset analysis
- Designed a PostgreSQL schema supporting multi-organization accounts for hydrogeologic and historical well data

LEADERSHIP & ACTIVITIES

Cornell Assistive Technologies Ithaca, NY
Website Team Lead & Software Engineer Oct 2025 - Present

- Led a 7-person team to revamp the organization's website from a Wix-based template into a custom full-stack application with a team-wide CMS
- Shipped the core website in 2 months, with the CMS used by 8 content admins across a 60-member team
- Developed an eye-tracking interface for an accessible electric wheelchair in Python, enabling control for users with limited mobility

SKILLS

Languages: TypeScript, Python, Java, C, RISC-V Assembly, Rust

Frontend: React, React Native, Next.js, Tailwind CSS, XState, Redux, Jest

Backend: Node.js (Express, Next.js API), REST, GraphQL

Cloud & Databases: AWS (Lambda, DynamoDB, Bedrock, Cognito), SQL, PostgreSQL, Vercel, Prisma

Machine Learning & Data Analytics: TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy

Version Control & CI/CD: Git, GitHub Actions, Jenkins, Docker